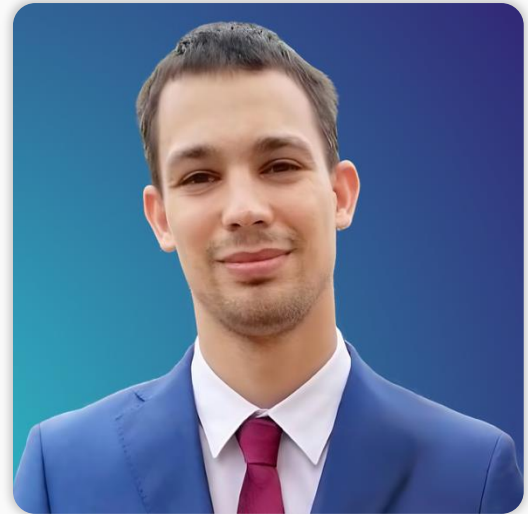


Flutter & AWS Amplify Gen2

Sviluppiamo un'applicazione fullstack multiplatforma a tempo record!



Federico

Parezzan

Senior Software Engineer

Java Oracle Certified

Flutter Developer



1. What is Flutter?

WHAT IS FLUTTER?

Flutter is Google's UI toolkit

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Flutter is Google's UI **framework**

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Flutter is Google's UI framework
for building **natively compiled** applications
for mobile, web, and desktop

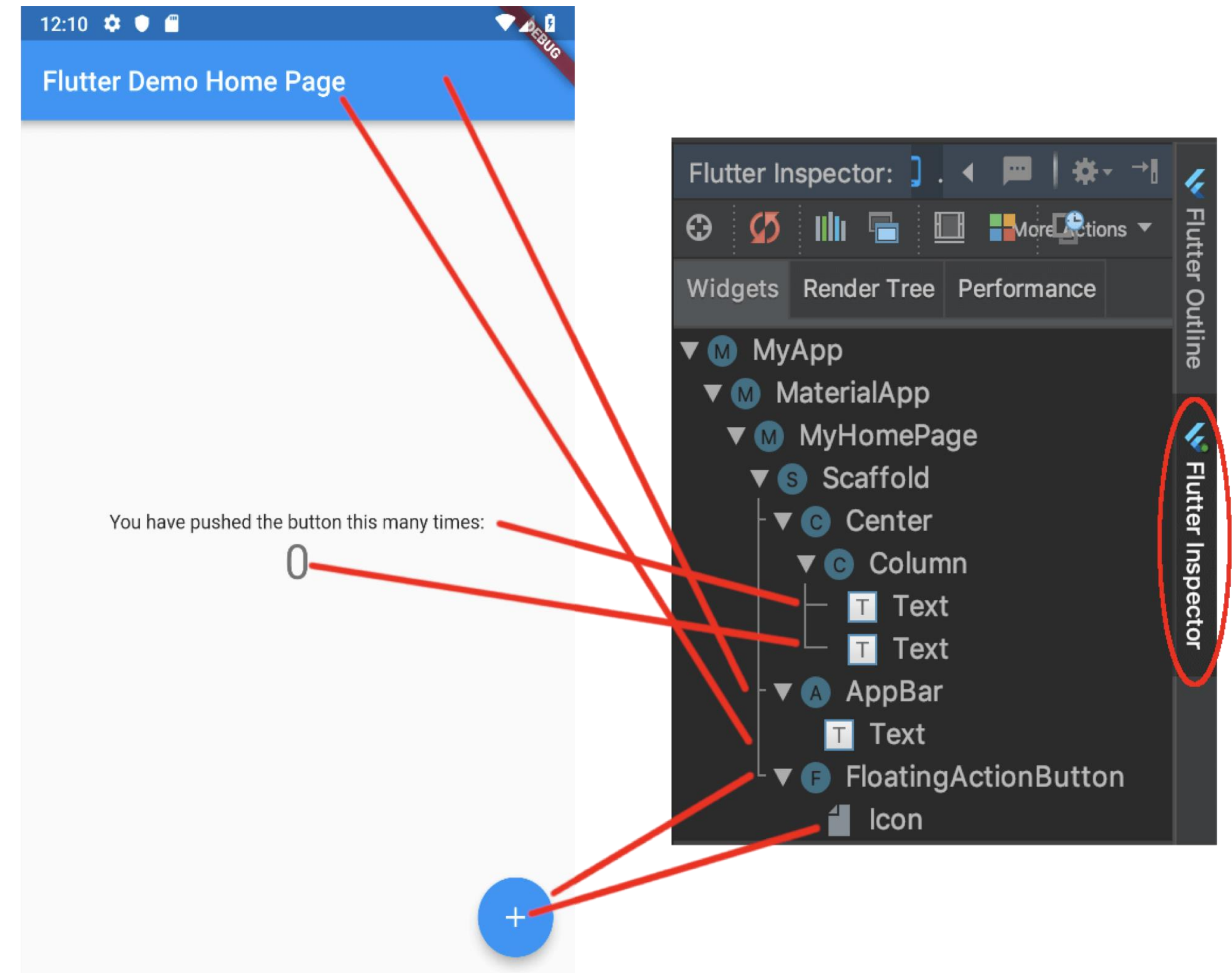
WHAT IS FLUTTER?

Flutter is Google's UI framework
for building natively compiled applications
for mobile, web, and desktop
from a **single codebase**.

WHAT IS FLUTTER?

Everything is a widget!

A **widget** defines a UI element based on its configuration and state.



WHAT IS FLUTTER?

Main types of widget

StatelessWidget

➡ No internal state

Main types of widget

StatelessWidget

- ➔ No internal state
- ➔ It is built once

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Main types of widget

StatelessWidget

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- ➔ It is built once
- ➔ It does not change

StatefulWidget

- ➔ Has a state that can change over time, triggering a rebuild.

2. What is Amplify?

WHAT IS AMPLIFY

AWS Amplify is everything Flutter developers need to develop **cloud-powered** fullstack applications without hassle.

WHAT IS AMPLIFY

Easily connect your Flutter applications to the cloud for:

1

Data modeling

2

Authentication

3

Storage

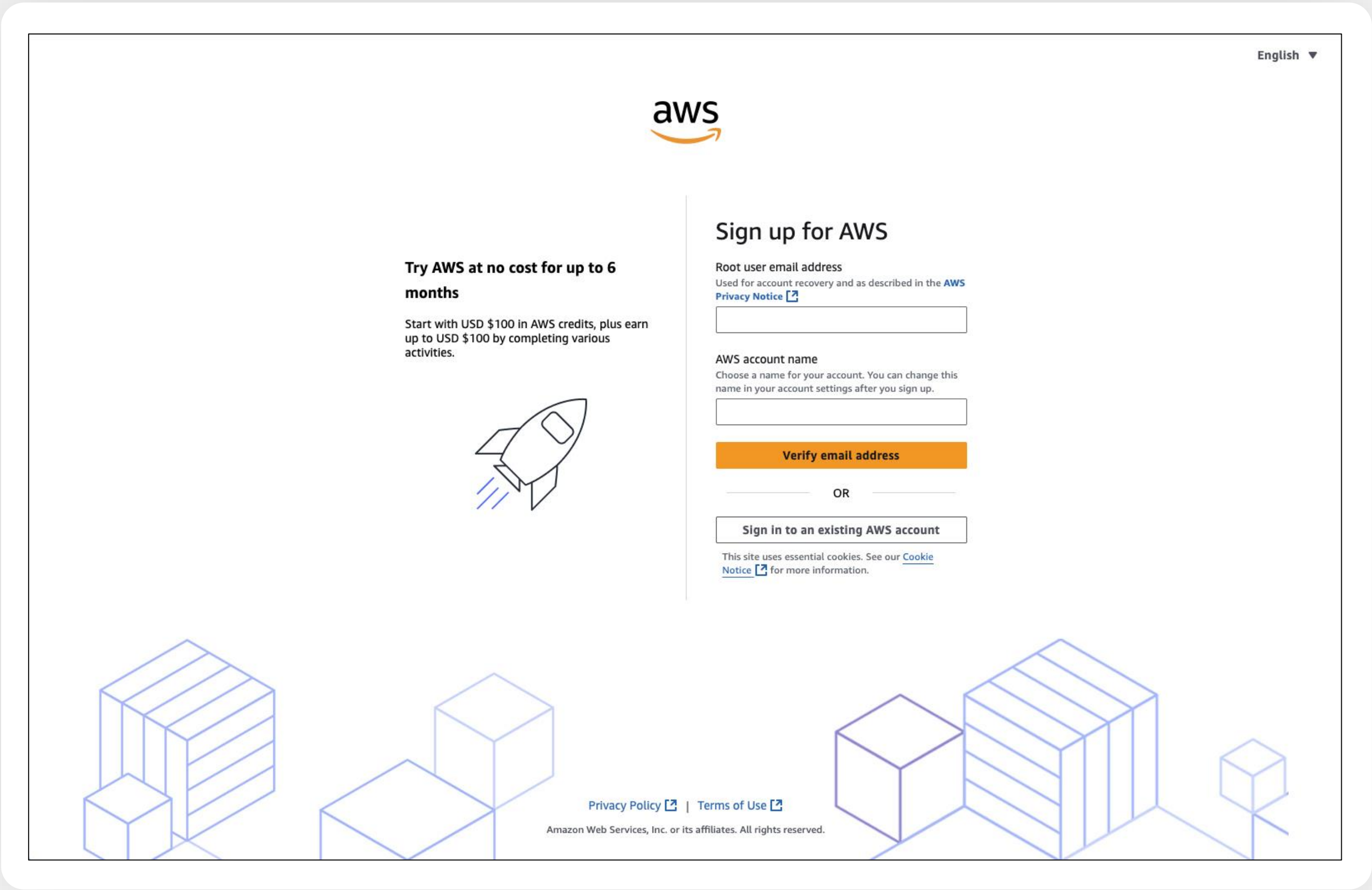
4

Serverless functions

and more...

3. Configuring the environment

Account aws

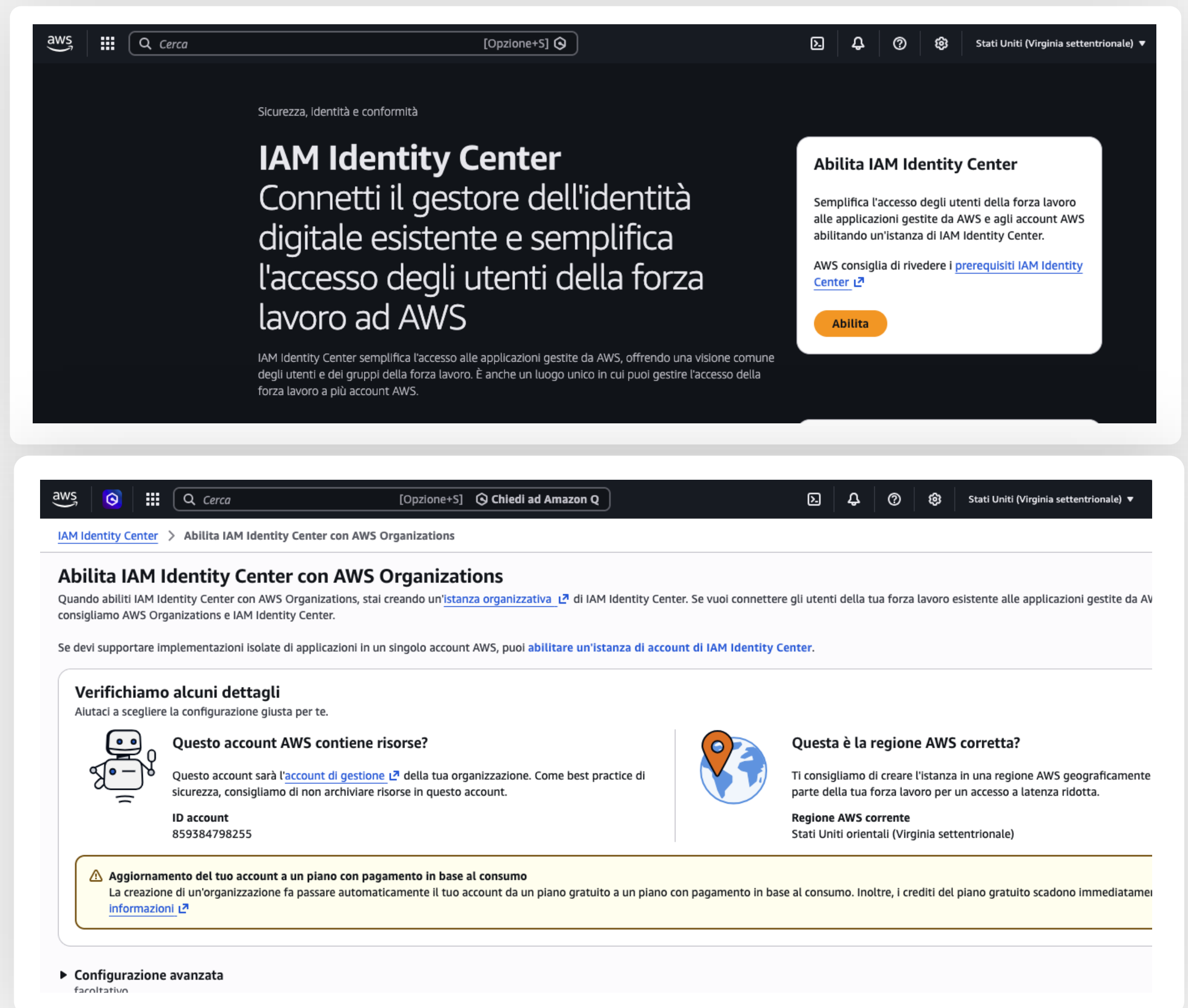


Create user with Amplify permissions

Sign in to the AWS Console to access **IAM Identity Center** page and choose Enable. Then set a password.

Suggested by documentation but...

⚠ **Aggiornamento del tuo account a un piano con pagamento in base al consumo**
La creazione di un'organizzazione fa passare automaticamente il tuo account da un piano gratuito a un piano con pagamento in base al consumo. Inoltre, i crediti del piano gratuito scadono immediatamente. [Ulteriori informazioni](#)



Create user with Amplify permissions

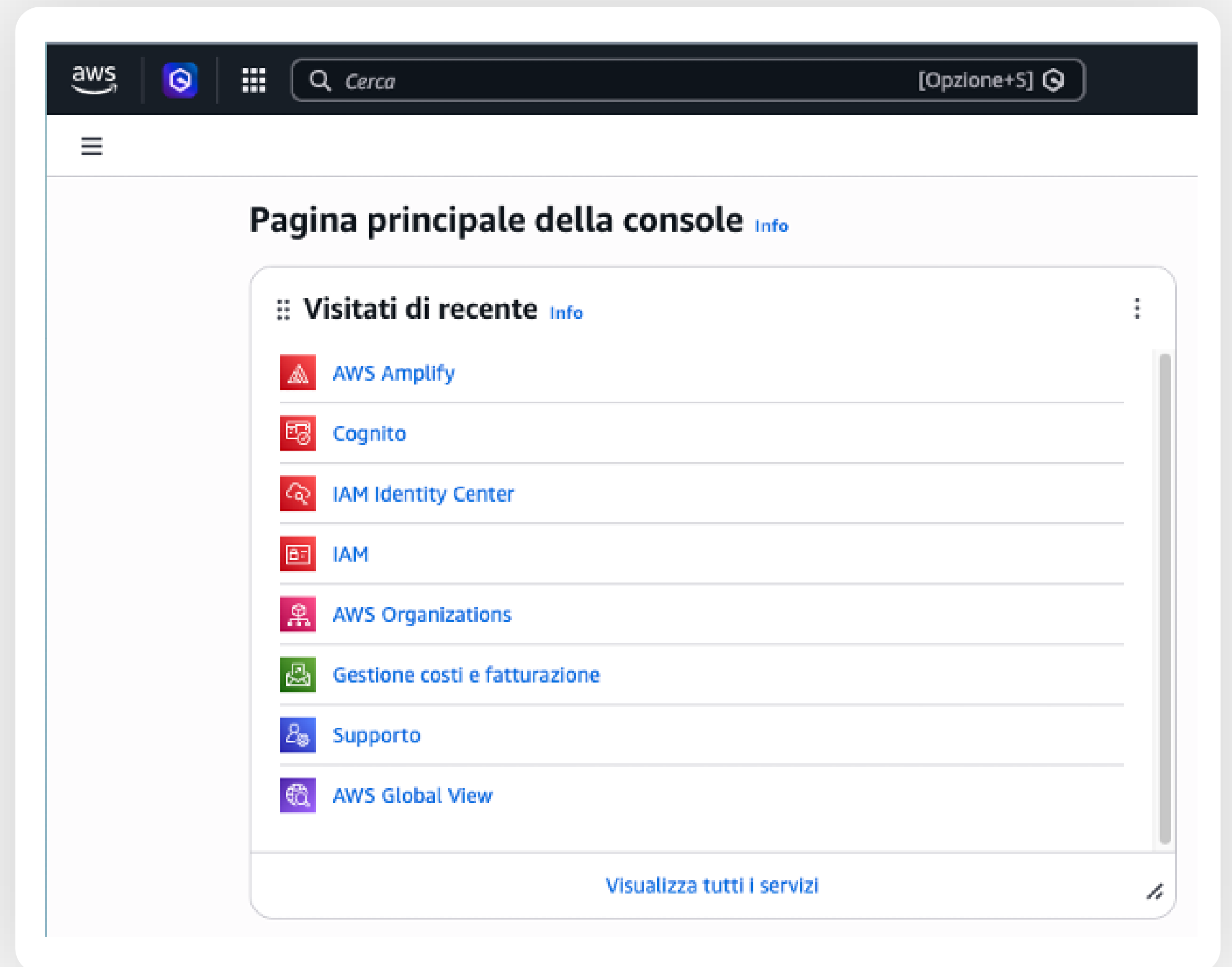
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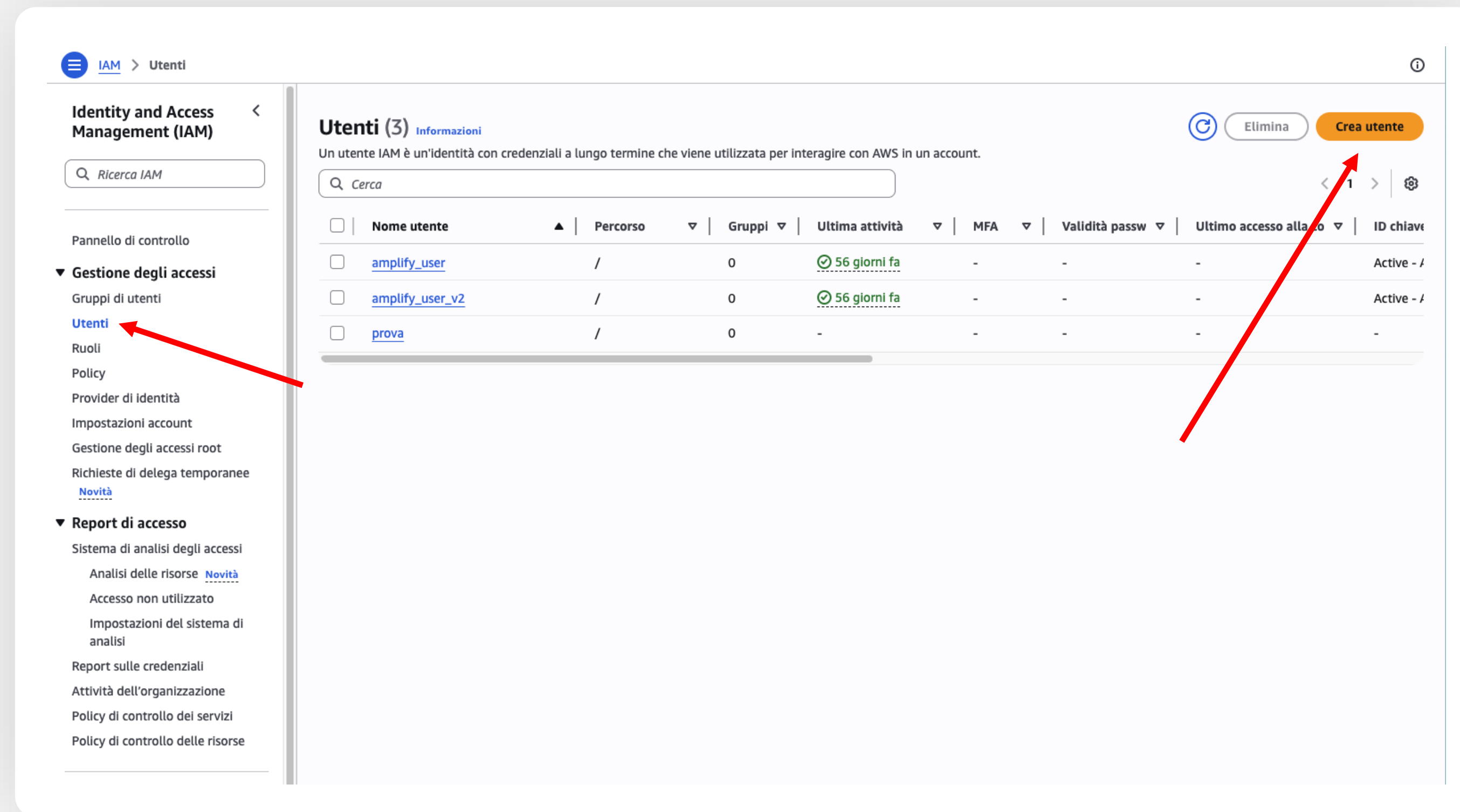


Create user with Amplify permissions (free is better)

Sign in to the AWS Console to
access **IAM** page

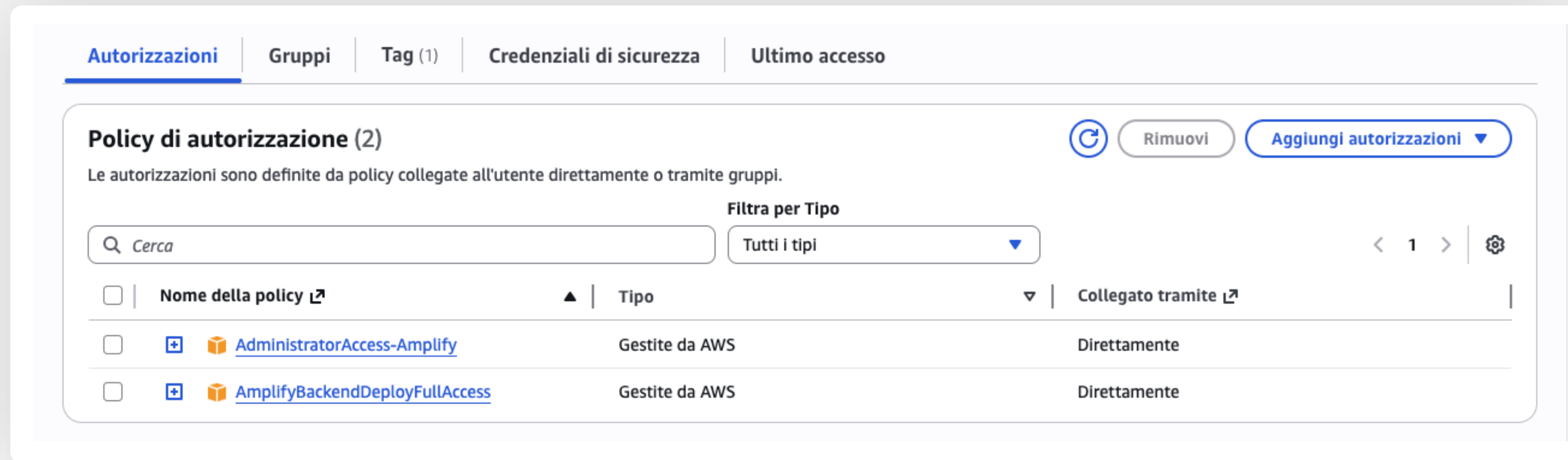


Create user with Amplify permissions (free is better)



Click **User** in the left panel and then the **Create user** button

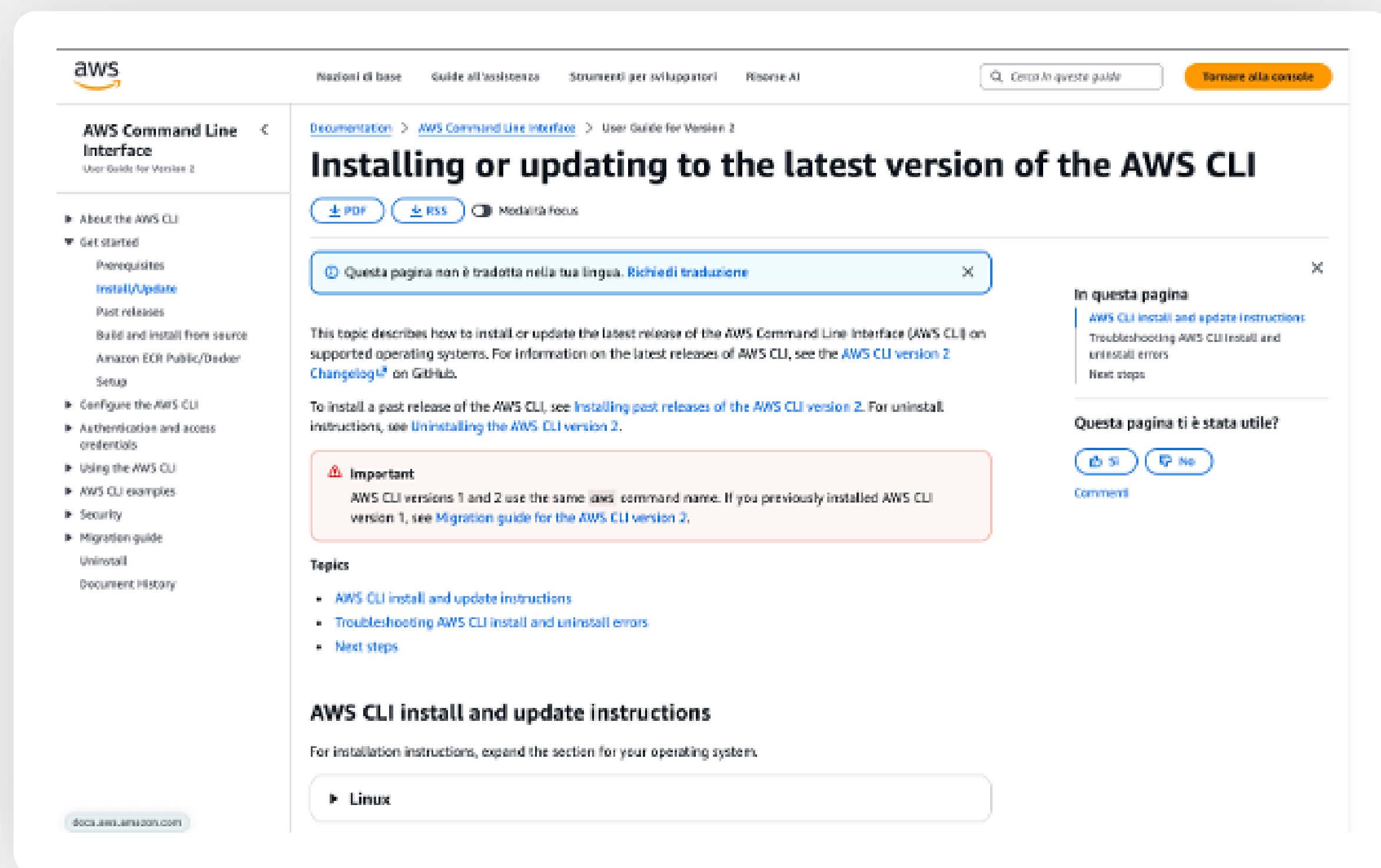
Create user with Amplify permissions (free is better)



Add the policies: **AmplifyBackendDeployFullAccess**
and **AdministratorAccess-Amplify**

CONFIGURING THE ENVIRONMENT

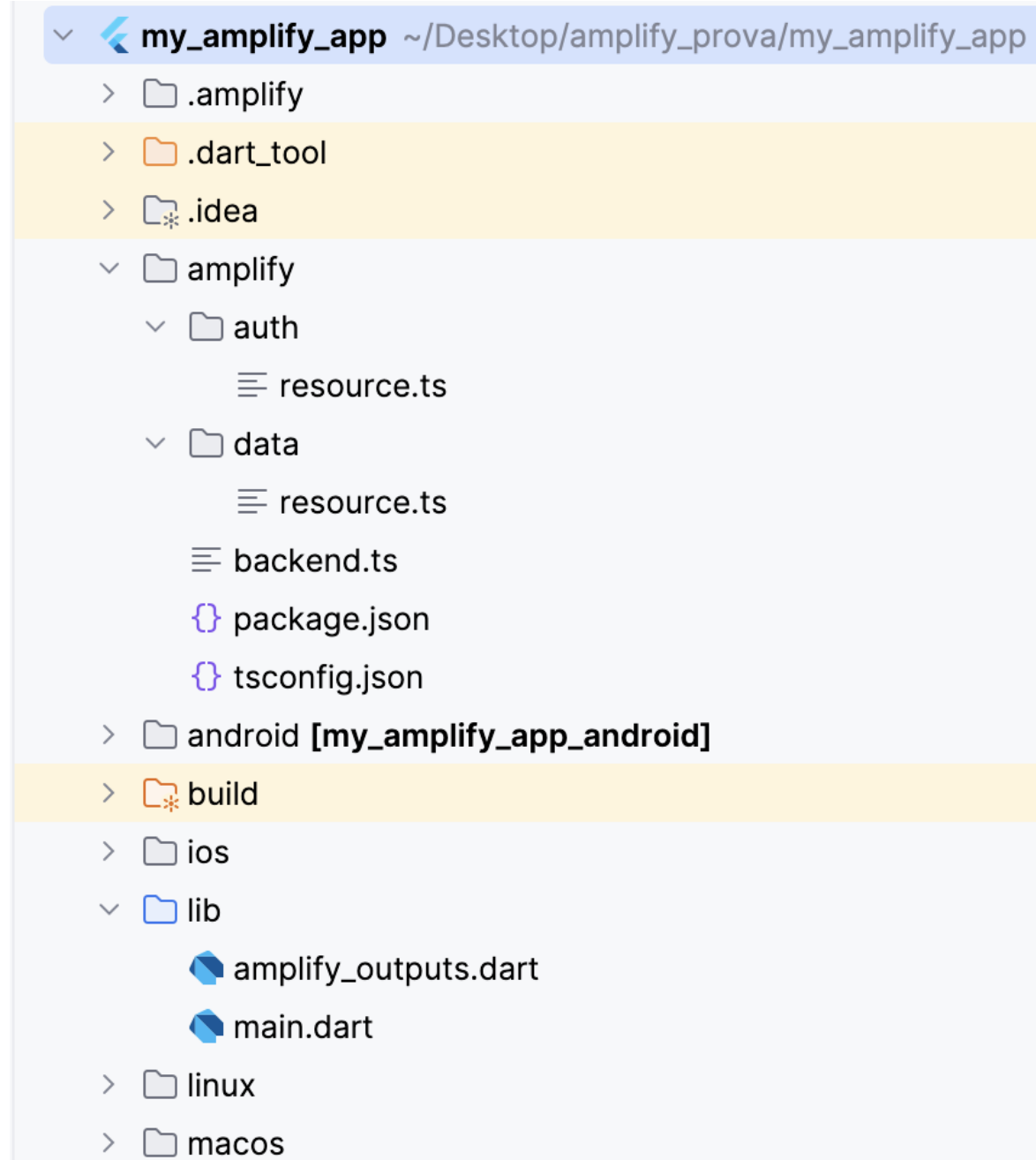
Install the AWS CLI



<https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>

Create AWS Amplify project

```
1 flutter create my_amplify_app
2 cd my_amplify_app
3 npm create amplify@latest -y
4 sudo amplify init
```



The screenshot shows a file explorer window for the project 'my_amplify_app' located at '~/Desktop/amplify_prova/my_amplify_app'. The project structure is as follows:

- > .amplify
- > .dart_tool
- > .idea
- ▼ amplify
 - ▼ auth
 - resource.ts
 - ▼ data
 - resource.ts
 - backend.ts
 - package.json
 - tsconfig.json
- > android [my_amplify_app_android]
- > build
- > ios
- ▼ lib
 - amplify_outputs.dart
 - main.dart
- > linux
- > macos

Set up local AWS profile

```
1 sudo aws configure --profile amplify_user_v2
```

IAM > Utenti

Identity and Access Management (IAM)

Ricerca IAM

Pannello di controllo

Gestione degli accessi

Gruppi di utenti

Utenti

Ruoli

Policy

Provider di identità

Impostazioni account

Gestione degli accessi root

Richieste di delega temporanee

Novità

Report di accesso

Sistema di analisi degli accessi

Analisi delle risorse

Novità

Accesso non utilizzato

Impostazioni del sistema di analisi

Report sulle credenziali

Attività dell'organizzazione

Policy di controllo dei servizi

Policy di controllo delle risorse

Utenti (3) Informazioni

Elimina

Crea utente

Un utente IAM è un'identità con credenziali a lungo termine che viene utilizzata per interagire con AWS in un account.

Cerca

	Nome utente	Percorso	Gruppi	Ultima attività	MFA	Validità passw	Ultimo accesso alla co	ID chiave
☐	amplify_user	/	0	56 giorni fa	-	-	-	Active - /
☐	amplify_user_v2	/	0	56 giorni fa	-	-	-	Active - /
☐	prova	/	0	-	-	-	-	-

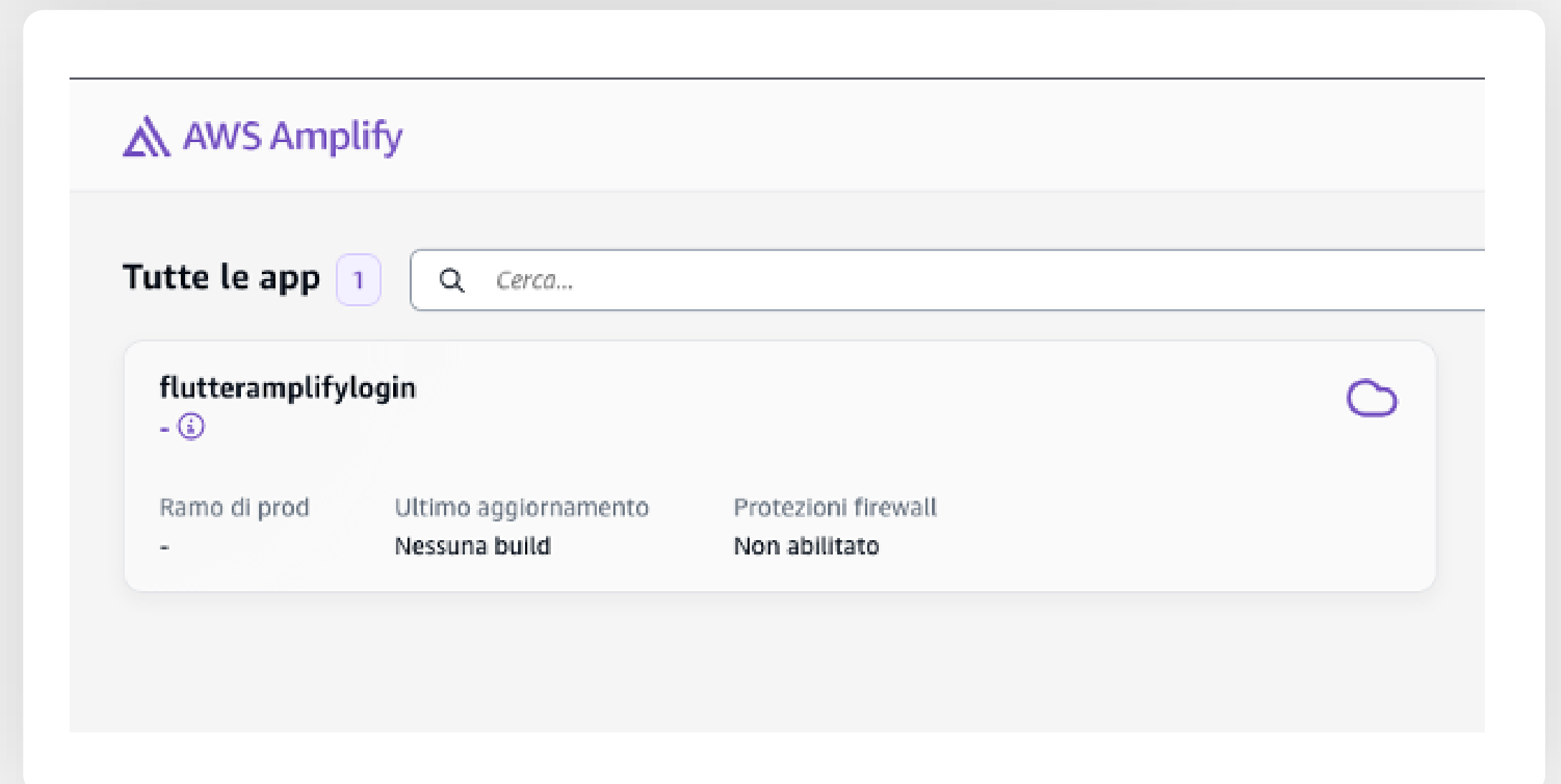
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Bootstrap your AWS account

```
1 sudo npx ampx sandbox --outputs-format dart --outputs-out-dir lib --profile amplify_user_v2
```

- Sign in to the AWS Console
- On the **Create new app** page, choose **Initialize setup now**



What is sandbox?

```
1 sudo npx ampx sandbox --outputs-format dart --outputs-out-dir lib --profile amplify_user_v2
```

An **isolated, temporary cloud environment** for dev.

- Each developer gets their own backend
- Real-time updates while coding
- No impact on dev/prod environments
- Perfect for testing and fast iteration

👉 **Think:** “a Git branch, but for your backend”

CONFIGURING THE ENVIRONMENT



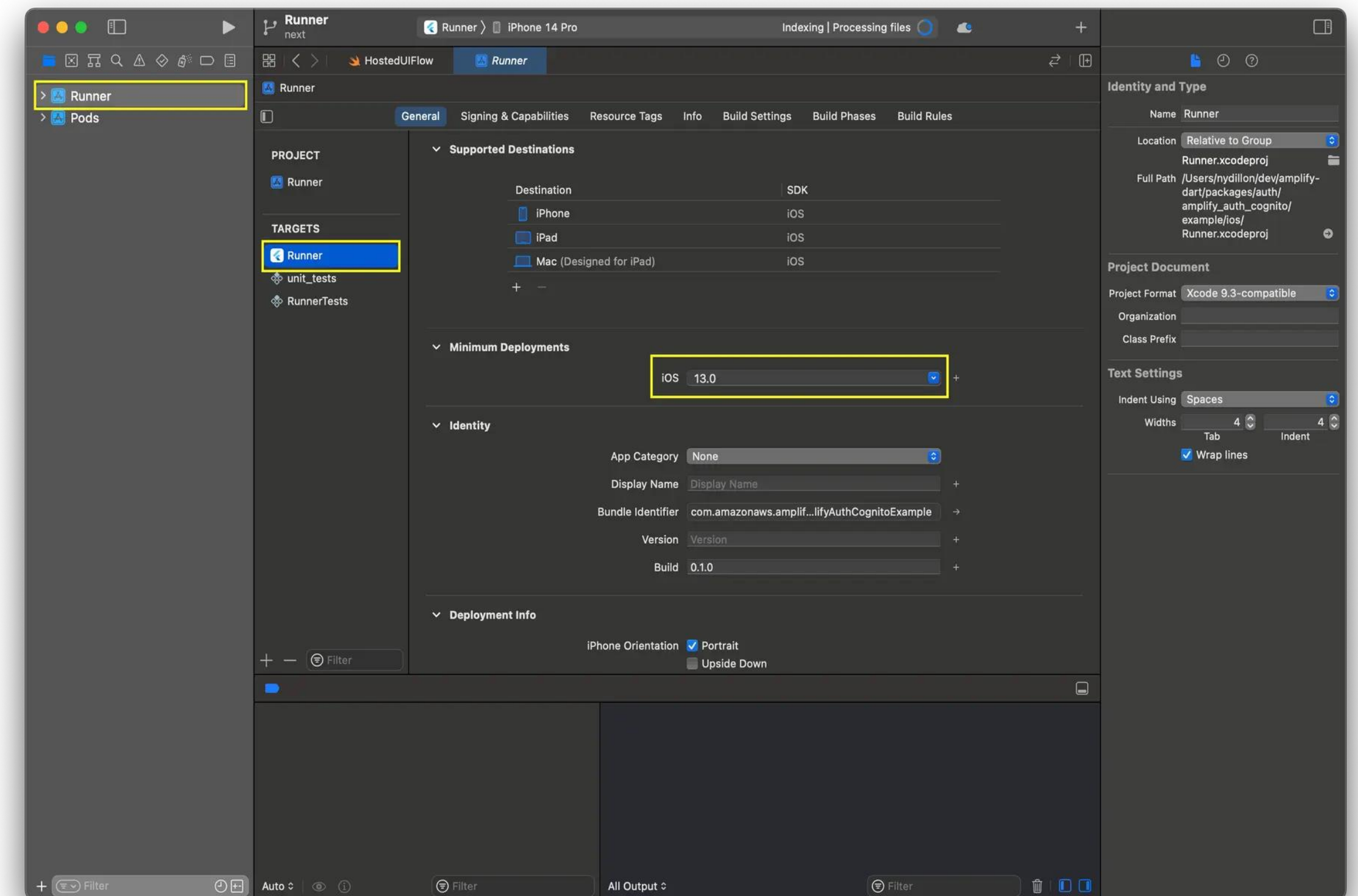
4. Configuring the platforms

CONFIGURING THE PLATFORMS

iOS setup

Open ios/Podfile and update the target iOS platform to 13.0 or higher.

Minimum deployment target of 13.0
Xcode 15.0



Android setup

Network Permissions for Release Builds

API level 24+ (Android 7.0+)
Gradle 8+, Kotlin 1.9+, Java 17+

```
1  android {
2      namespace = "com.example.myapp"
3      compileSdk = flutter.compileSdkVersion
4      ndkVersion = flutter.ndkVersion
5      compileOptions {
6 -         sourceCompatibility = JavaVersion.VERSION_1_8
7 -         targetCompatibility = JavaVersion.VERSION_1_8
8 +         sourceCompatibility = JavaVersion.VERSION_17
9 +         targetCompatibility = JavaVersion.VERSION_17
10     }
11
12     kotlinOptions {
13 -         jvmTarget = JavaVersion.VERSION_11.toString()
14 +         jvmTarget = JavaVersion.VERSION_17.toString()
15     }
16
17     defaultConfig {
18         // TODO: Specify your own unique Application ID
19         (https://developer.android.com/studio/build/application-id.html).
20         applicationId = "com.example.myapp"
21         // You can update the following values to match your application needs.
22         // For more information, see:
23         https://docs.flutter.dev/deployment/android#reviewing-the-gradle-build-configuration.
24 -         minSdk = flutter.minSdkVersion
25 +         minSdk = 24
26         targetSdk = flutter.targetSdkVersion
27         versionCode = flutterVersionCode.toInteger()
28         versionName = flutterVersionName
29     }
30
31     buildTypes {
32         release {
33             // TODO: Add your own signing config for the release build.
34             // Signing with the debug keys for now, so `flutter run --release` works.
35             signingConfig = signingConfigs.debug
36         }
37     }
38 }
```

WEB setup

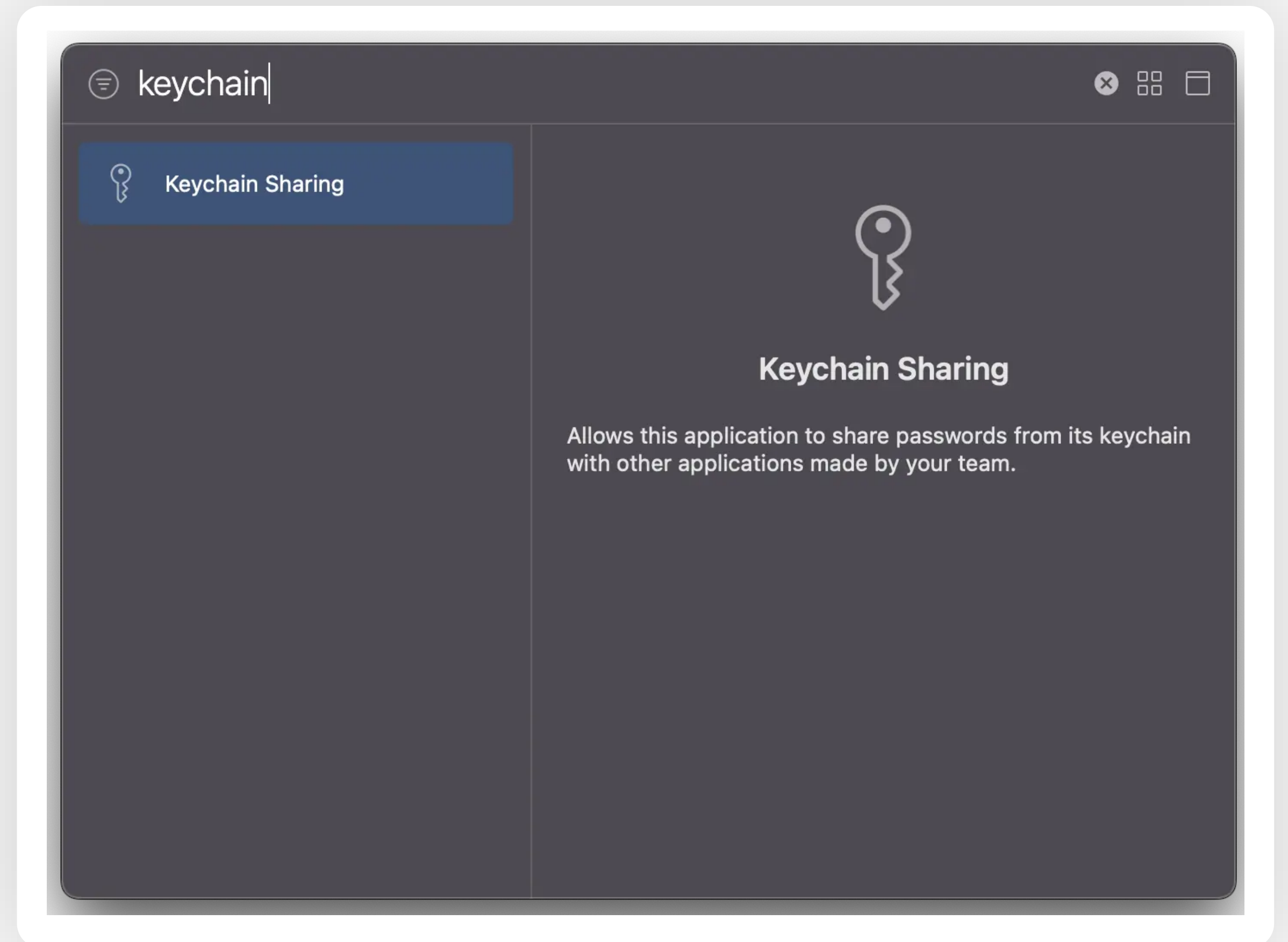


CONFIGURING THE PLATFORMS

macOS setup

Enable networking, keychain entitlements, and code signing.

minimum deployment target of 10.15
Xcode 15.0 or higher



Windows setup



Linux setup



```
1 sudo apt-get update
2 sudo apt-get install -y libsecret-1-dev
```

Amplify Flutter depends on the libsecret library

5. Demo

**ONE FRAMEWORK FOR UI...
ONE BACKEND TO BIND THEM...**

NON SO SCRIVERE LE POESIE

USA FLUTTER + AMPLIFY

Federico Parezzan

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touch:

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You can
find me:

[Portfolio](#)

[Linkedin](#)

[Github](#)